

AdInsight: A Persuasive Advertising Explainer

DATA/MSAI/MSML 641: Natural Language Processing

Group 6 — Final Project Report

1. Problem and Users

Online advertisements often use persuasive language designed to influence users to make quick decisions. Techniques such as urgency, scarcity, fear appeals, social proof, exaggerated claims, and appeals to authority can make an advertisement feel more trustworthy or time-sensitive than it actually is. These techniques are not necessarily deceptive on their own, but they can make it more difficult for users to slow down and evaluate what an advertisement is asking them to believe or do.

Our project, AdInsight, was designed to make these persuasion techniques easier to recognize. Rather than labeling an advertisement as good, bad, legitimate, or a scam, the goal is to show users how the language in an advertisement is attempting to persuade them. The system translates model predictions into plain-language explanations that do not require users to understand NLP or machine-learning terminology.

Our primary target population is adults age 60 and older. We chose this group because older adults regularly encounter online advertisements for products and services such as health products, supplements, insurance, and financial services. These advertisements may use artificial scarcity, pseudo-scientific language, emotional appeals, and other persuasive techniques that are not always easy to recognize while browsing online. However, AdInsight can also be useful for younger consumers who encounter influencer marketing, wellness claims, financial advertisements, and FOMO-based promotions.

Without a tool like AdInsight, a user would typically have to interpret an advertisement on their own or independently research its claims. This can be especially difficult when the concern is not a clearly false statement, but rather the way information is framed. For example, phrases such as “only three left,” “doctor recommended,” or “offer ends tonight” may influence a user’s decision without necessarily containing an explicitly false statement. Our labeling guidelines therefore focus on the persuasive language itself rather than attempting to classify advertisements as scams or illegal content.

The main value of AdInsight is giving users a reason to slow down and look more closely at how an advertisement is trying to influence them. Instead of making the decision for the user, the system identifies persuasive techniques and provides information that can help the user evaluate the advertisement more critically.

2. Product

AdInsight is an end-to-end advertising analysis system that accepts an advertisement and returns a plain-language explanation of the persuasive tactics it detects. The final system supports both text and image-based advertisements and can be accessed through a desktop/web application or a Chrome browser extension.

The process begins when a user encounters an advertisement they want to analyze. In the application, the user can paste advertisement text directly or upload a screenshot. The browser extension allows the user to analyze an advertisement while viewing the webpage where they encountered it. Both interfaces connect to the same backend, giving users multiple ways to access the same underlying analysis.

For text input, the advertisement can be passed directly into the NLP pipeline. Image advertisements require an additional extraction step to convert the text in the screenshot into a form the classifier can process. The project initially relied on RapidOCR for this step, but later testing showed that complex advertisements could cause OCR to lose or incorrectly combine important words. The final system uses a vision-language model (VLM) as the primary transcription method when available, with RapidOCR retained as a fallback if the VLM is unavailable or fails. The VLM is intentionally restricted to transcription: it extracts what the advertisement says but does not determine which persuasive tactic is being used. That classification remains the responsibility of the fine-tuned DistilBERT model.

After the text is extracted, the classifier predicts a persuasion category. The system also compares the text with trigger phrases defined in our labeling guidelines. These two signals are shown separately in the interface. A classifier-based

result is identified as the model’s prediction, while a phrase directly matched in the advertisement is identified as a phrase match. This distinction is important because the classifier is single label, while a real advertisement may contain more than one persuasive technique.

AdInsight also includes a review-insight layer that provides additional context about the product being advertised. When a product or company can be identified, the system searches public web sources for relevant information and ranks sources based on their independence from the advertiser. Advertiser-owned pages are excluded, while potential affiliate content is flagged and ranked below independent sources. The persuasion tactic detected by the classifier also helps shape the search strategy; for example, exaggerated or authority-based claims prompt searches focused on evidence, while urgency, scarcity, and FOMO can prompt searches related to subscription, cancellation, or refund experiences.

The review layer presents this information through plain-language definitions of technical or authoritative-sounding terms, a short summary grounded only in retrieved passages, and links to the underlying sources so users can review the information themselves. If no identifiable product or company is found, the system reports that there is nothing to look up rather than attempting to guess. If the review service is unavailable or cannot retrieve usable results, the interface falls back to clearly labeled sample data rather than interrupting the main ad analysis.

The final interface turns these results into a short explanation instead of showing the user only probabilities or model labels. For image advertisements, the application also displays the text that was extracted before classification. This gives users a way to verify what the system analyzed and helps make possible OCR or transcription errors more transparent. Overall, AdInsight takes text or image-based advertisements, extracts the relevant text when needed, analyzes it with the DistilBERT classifier and phrase-matching system, and returns a plain-language explanation along with outside context from the review-insight layer.

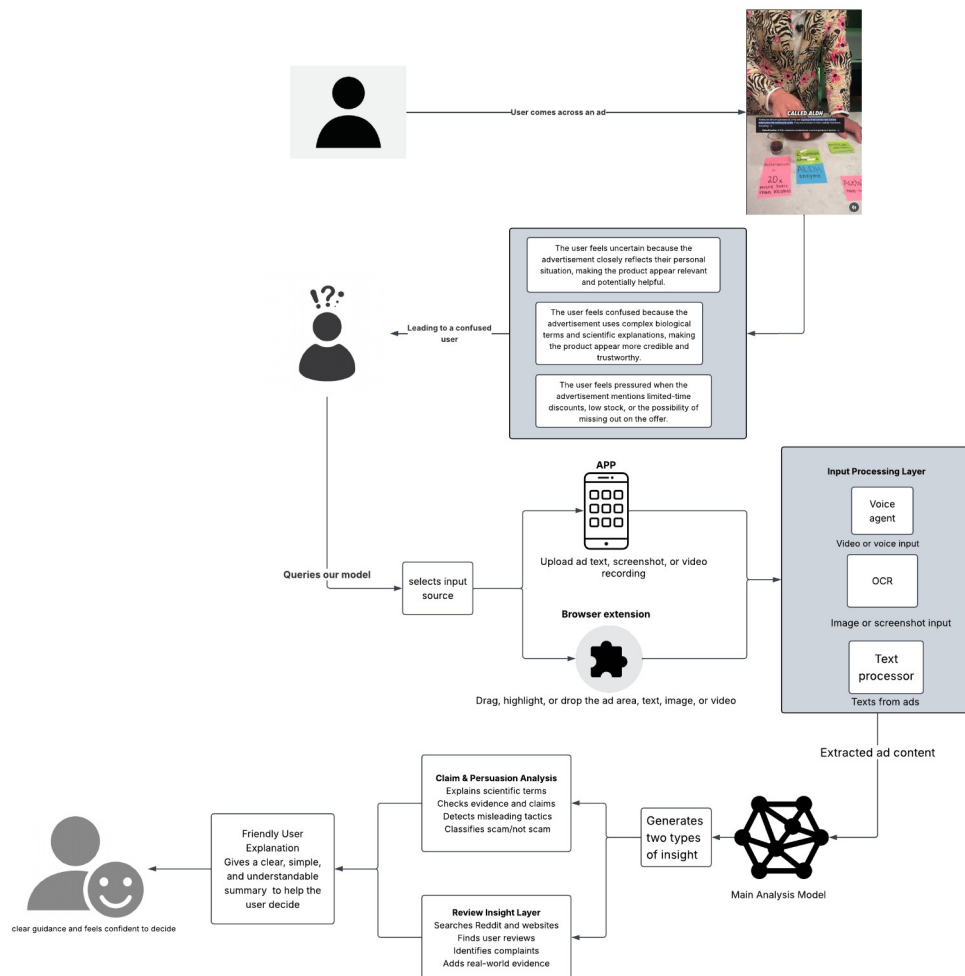


Figure 1: AdInsight end-to-end system — ad input (app or browser extension), text extraction, analysis and review insight, and plain-language explanation.

3. NLP Method and Evaluation

Data

Our NLP task is a persuasion-tactic classification problem. The final merged dataset contains 4,374 advertisements across eight classes: Urgency, Exaggerated Claims, Scarcity, Social Proof, Authority Manipulation, Fear Appeals, Neutral, and FOMO.

Class	Examples
Urgency	959
Exaggerated Claims	905
Scarcity	798
Social Proof	510
Authority Manipulation	389
Fear Appeals	304
Neutral	287
FOMO	222
Total	4,374

The project originally used a smaller seven-class dataset. Analysis of that dataset revealed that some classes contained highly repetitive phrases, which created a risk that the model could learn shortcuts rather than broader persuasive patterns. For example, “limited stock” appeared in 435 of 449 Scarcity examples. To increase the variety of language in the training data, we incorporated 857 examples from the Mathur et al. dark-patterns corpus, which contains real-world e-commerce language collected from shopping websites. Compatible examples were mapped to our existing Scarcity, Urgency, and Social Proof categories.

We also added a Neutral category because the earlier dataset contained only advertisements associated with a persuasion tactic. Without Neutral examples, the classifier was effectively forced to select a persuasion category even when no clear tactic was present. After these changes, the final training corpus contained 4,374 examples across eight classes.

The dataset remained imbalanced, with 959 Urgency examples compared with 222 FOMO examples. To reduce the impact of this imbalance during training, we used class-weighted cross-entropy loss, giving less frequent classes greater weight in the loss calculation.

Model and Training

We selected DistilBERT as the primary NLP classifier and fine-tuned the pretrained distilbert-base-uncased model for our persuasion-tactic classification task.

The data was divided using a stratified 70/15/15 train-validation-test split so that each subset maintained approximately the same class distribution. The test set remained held out during model selection and hyperparameter tuning.

Our primary evaluation metric was macro F1. Because the dataset is imbalanced, accuracy alone could hide weaker performance on the less common persuasion categories. Macro F1 calculates the F1 score for each class independently and gives each class equal weight, making it a better measure of performance across all eight categories.

To provide baselines for comparison, we also evaluated two non-neural approaches using the same stratified data split as the DistilBERT model. The first was a majority-class baseline that always predicted Urgency, the most frequent class in the dataset. The second used TF-IDF word unigram and bigram features with class-balanced logistic regression. These baselines provide both a simple lower bound and a stronger traditional NLP comparison for evaluating whether the transformer-based model provides additional value.

We first trained a baseline DistilBERT classifier and then used Hyperopt with Tree-structured Parzen Estimator (TPE) search to tune the training configuration. The search included learning rates of 1e-5, 2e-5, 3e-5, and 5e-5; batch sizes of 8 and 16; two, three, or four epochs; and weight-decay values of 0, 0.01, and 0.05. TPE sampled three trial configurations from this space, selected on validation macro F1.

Selected hyperparameters:

- Learning rate: 5e-5
- Batch size: 16
- Epochs: 3
- Weight decay: 0.0

The best configuration was then retrained before being evaluated on the held-out test set.

Results

The final eight-class model achieved a test accuracy of 0.9285 and a macro F1 of 0.9191.

Metric	Majority Class	TF-IDF + Logistic Regression	Final DistilBERT
Test Accuracy	0.2192	0.9011	0.9285
Test Macro F1	0.0449	0.8946	0.9191

The majority-class baseline performed poorly, with 0.2192 accuracy and 0.0449 macro F1, showing that simply predicting the most common category does not provide useful performance across the eight persuasion classes. The stronger TF-IDF and logistic regression baseline achieved 0.9011 accuracy and 0.8946 macro F1. The final DistilBERT model improved these results to 0.9285 accuracy and 0.9191 macro F1. This comparison shows that traditional lexical features captured much of the signal in the dataset, while the transformer-based model provided an additional improvement in performance across the persuasion categories.

The earlier seven-class model achieved a test accuracy of 0.9090 and macro F1 of 0.8960. After expanding the dataset and adding the Neutral class, the final model improved to 0.9285 accuracy and 0.9191 macro F1 across all eight classes. When evaluating only the seven persuasion categories represented in the earlier model, macro F1 was 0.9070. Social Proof also showed improvement, increasing from an F1 of 0.77 to 0.88.

While these held-out test results were strong, they did not tell the complete story. We later evaluated the full system on six genuine advertisement screenshots collected from X, Instagram, and short-form video. These advertisements were separate from the training corpus and gave us a more realistic view of how the system performed on advertising encountered online.

We first compared text-extraction methods across 53 required strings contained in the six advertisements. Raw RapidOCR recovered 55% of the required text, our improved OCR pipeline recovered 64%, and the VLM recovered 89%. This showed that the VLM was substantially better at extracting text from more complex advertisement layouts.

However, better text extraction did not result in better classification. Using RapidOCR or the application's OCR pipeline, the correct persuasion tactic was identified for only one of the six advertisements. Using the VLM-extracted text, none of the six advertisements were classified correctly. Four of the advertisements contained income-related claims, a type of advertising that was not represented in the training data or trigger-phrase lexicon.

This real-world evaluation revealed an important limitation that was not as visible in the held-out test results. Although the model performed well on data like its training distribution, it had difficulty generalizing to types of advertisements that were not well represented in the dataset. This suggests that the next major improvement to AdInsight should focus on expanding the diversity of the training data rather than only improving text extraction.

4. User Evidence

Our validation used six users between ages 60 and 74. Each user completed four tasks: analyzing a pasted text advertisement, analyzing an advertisement screenshot, identifying the main persuasion tactic shown in the result, and explaining what the result meant in their own words.

Overall, users were able to navigate the main AdInsight workflow successfully. Five of the six users completed the text-ad task without assistance, and four of six completed the image-ad task independently. All six users were able to identify

the main persuasion tactic once the result was displayed. This suggests that the core purpose of the tool was clear and that users were able to understand the main output without needing technical knowledge.

Users also responded positively to the plain-language explanations provided by AdInsight. Five of the six users rated the explanation as useful for understanding how the advertisement was trying to influence them. The ability to connect the result back to specific wording in the advertisement was especially helpful because it made the analysis feel more transparent and easier to follow.

The image-analysis portion also received positive feedback. Users appreciated being able to see the text that AdInsight extracted from the advertisement before the result was shown. This gave them a way to confirm what the system analyzed and helped make the process easier to understand.

The validation also showed that most users were comfortable with the overall layout and flow of the product. The main areas for future improvement were related to terminology and uncertainty rather than the core design. Some users were less familiar with terms such as “model pick,” and low-confidence results could benefit from additional explanation, so users understand that the absence of a strong prediction does not mean an advertisement has been verified as safe or trustworthy.

Overall, the validation supported the direction of AdInsight and showed that the main workflow was understandable to the target audience. Users were able to complete the primary tasks, understand the main persuasion result, and see value in receiving a plain-language explanation rather than only a model label or confidence score.

Future user testing could also examine how users interact with the review-insight layer, including whether the outside-source summaries, source links, and plain-language definitions provide useful context alongside the persuasion analysis.

In future versions, we would continue improving the clarity of the interface by simplifying model-related terminology, making low-confidence results easier to interpret, and keeping the extracted advertisement text visible so users can clearly see what AdInsight analyzed. Additional testing with a larger and more diverse group of older adults would also help determine whether these findings remain consistent across different levels of digital experience.

5. Ethics and Limitations

AdInsight analyzes content that can influence financial, health, and purchasing decisions, so the system should not present itself as an authority on whether an advertisement is truthful, legal, or safe. Our labeling framework is intended for educational NLP analysis rather than legal classification, and advertisements are not labeled as scams or illegal content.

One major limitation is bias in the training data. Although the final dataset is larger and more diverse than the original corpus, it still does not represent every type of advertising users may encounter. This became clear during the real-ad evaluation, where several income-related advertisements were poorly classified because that type of content was not well represented in the training data. Strong held-out test performance therefore should not be interpreted as evidence that the model will perform equally well on all real-world advertisements.

Text extraction is another limitation. OCR can lose or combine words when advertisements use complex layouts, overlapping images, or unusual formatting. In our evaluation, the VLM recovered more of the required text than the OCR pipeline, but it also introduces an external network dependency. Its role is therefore restricted to transcription, while persuasion classification remains the responsibility of the fine-tuned DistilBERT model.

Privacy is also an important consideration because users may upload screenshots containing information beyond the advertisement itself. Future versions should minimize unnecessary data processing and clearly communicate what information is analyzed, whether image data is sent to an external service, and whether submitted content is retained.

The review-insight layer introduces additional external dependencies because web retrieval and language-model processing rely on outside services. The retrieved information should be treated as context rather than verification, since public web sources may contain opinions, incomplete information, or inaccurate claims. To reduce this risk, AdInsight excludes advertiser-owned pages, flags potential affiliate content, prioritizes more independent sources, and restricts its generated summary to information contained in the retrieved passages.

Finally, AdInsight should support rather than replace user judgment. The real-ad evaluation showed that the classifier can still produce incorrect predictions, including high-confidence errors. For this reason, AdInsight is best positioned as

an educational tool that highlights persuasive language and provides context without telling users definitively whether an advertisement or product can be trusted.